

Hand-lettering *the inscription.*

A brief for Jill: writing the master for Much Ado About One Side

*Come, palindove, let edges twine;
thy side is mine, and mine is thine.*

01 Overview

One poem, written once by hand in a slanted script, becomes the endless inscription on both foil editions of the sculpture.

02 Step by step

Twelve steps, each with a check you can do before moving on. Most of the time goes into warming up and writing; the rest is checking.

03 Templates and reference render

Ruled sheets at the working size, ghosted reference rows rendered in the stand-in font, the alphabet, the hard words, and a true-size preview of the ribbon.

04 The easy parts

If you only read one page, read this one. Sizes, the stroke rule, the checklist, and what to send back.

05 In depth

Proportions, spacing and joins, flourish zones, what the laser can and cannot mark, materials in detail, how the scan becomes foil, and what to do when things go wrong.

23 master rows · 7 mm x-height · 55° slant · thinnest stroke 1 mm · print every template sheet at 100 % and check its 100 mm bar.

The rendered script in this guide is the substitute font Great Vibes (SIL Open Font License), shown for size and spacing only. Project text and artwork: MIT OR Apache-2.0. muchadoaboutoneside.com/calligraphy

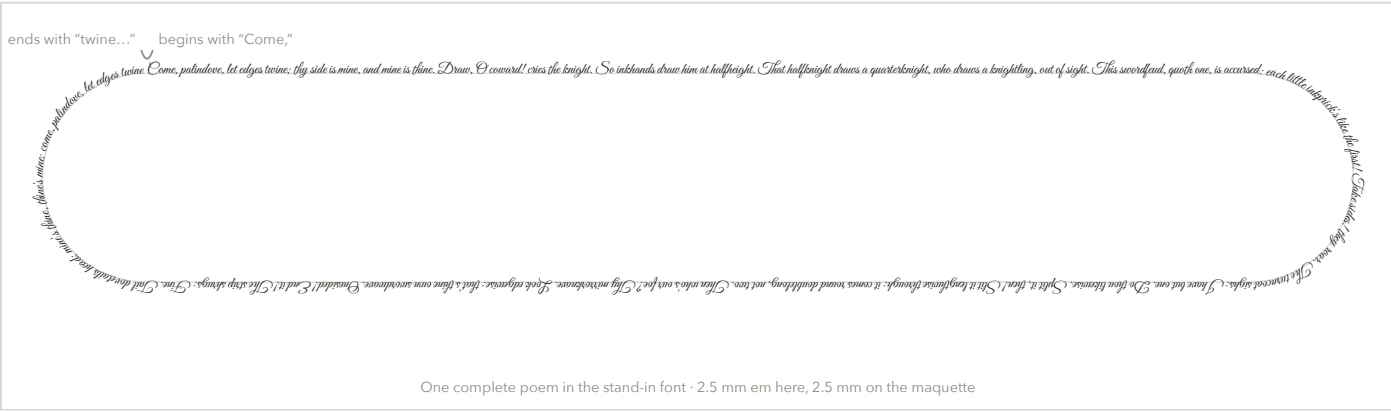
Overview

One poem, written once by hand in a slanted script, becomes the endless inscription on both foil editions of the sculpture.

The sculpture is a figure-eight snake wrapped in aluminum foil. The poem “Much Ado About One Side” is laser-marked onto that foil as one continuous ribbon of script that runs along the body, around the nose, back along the other side, around the tail, and straight into its own first word. The website currently shows the poem in a substitute font called Great Vibes. Your hand-lettering replaces it.

We do not need you to letter the sculpture or the foil. We need a master: the complete poem written on paper at a comfortable size, with consistent guidelines, so it can be scanned, traced into vector outlines, and scaled to each edition. The master is written once and reused everywhere.

The ribbon has no end. The final ellipsis after “twine...” runs straight into the opening “Come,”.



Where the lettering goes

EDITION	SIZE OF THE LETTERING	HOW THE POEM REPEATS
180 mm printed maquette (first physical run)	2.5 mm em, x-height about 0.89 mm	2 complete repetitions along one 1028 mm ribbon around the body
Full-size sculpture (conceptual study)	Set after fit tests; the same outlines scale up	4 rows around the body, each one complete poem on a closed reading path; the jaw carries its own 18-row inscription

Because the same outlines serve both editions, the master is drawn at one working size and every number in this guide is given in millimetres on your paper. The tables also show what each size becomes on the small maquette, which is the tightest case.

The deliverable in one sentence

23 labelled rows of black script on white paper, one poem line (or half line) per row, 7 mm x-height, 55° slant, no stroke thinner than 1 mm, scanned at 600 dpi.

Your hand, not the font

Great Vibes was chosen because its slanted script and flourished capitals resembled your sample. Treat the rendered reference sheets as a size and proportion guide only. Your letterforms, joins, and flourishes are what we want.

The poem, exactly

01 Come, palindove, let edges twine;
02 thy side is mine, and mine is thine.

03 Draw, O coward! cries the knight.
04 So inkhands draw him at halfheight.
05 That halfknight draws a quarterknight,
06 who draws a knightling, out of sight.
07 This swordfeud, quoth one, is accursed:
08 each little inkprick's like the first!
09 Take sides! they roar. The turncoat sighs:
10 I have but one. Do thou likewise.

11 Split it, then! Slit it lengthwise through:
12 it comes round doublelong, not two.
13 Then who's our foe? Thy mirrorknave.
14 Look edgewise: that's thine own swordwave.

15 Onesided! End it! The strip shrugs: Fine.
16 Tail dovetails head; mine's thine, thine's mine:
17 come, palindove, let edges twine...

❧ THE ELLIPSIS RUNS STRAIGHT INTO LINE 01

Step by step

Twelve steps, each with a check you can do before moving on. Most of the time goes into warming up and writing; the rest is checking.

01 Read the poem aloud from the exact-text sheet

Use the “The poem, exactly” page. Note the invented compound words (palindove, inkhands, halfheight, halfknight, quarterknight, knightling, swordfeud, doublelong, mirrorknave, swordwave), the apostrophes, and the single lowercase start of the last line. The wording is checked letter by letter against the source text.

CHECK You can name every punctuation mark on the inventory table without looking.

02 Gather materials

One pen (brush pen recommended), its black pigment cartridge, smooth bright-white paper, ruler, 2H pencil, eraser, low-tack tape, and a scrap of the same paper for pen tests. The materials table lists specific products and what to avoid.

CHECK Test-write “palindove” on the scrap; the ink dries matte black with no feathering.

03 Print the template sheets at 100 %

Print the eight ghosted sheets and at least ten blank ruled sheets. In the print dialog choose “Actual size” or 100 %, landscape, no “fit to page”. Print on the writing paper itself if the printer accepts it; otherwise print on plain paper and use it under your paper on a light pad.

CHECK Measure the bar in each sheet's header: it must be 100 mm, within 1 mm.

04 Warm up on the alphabet sheet

Write the lowercase alphabet twice and the capitals once at the working size, joined as you would in running text. Aim the body of a, o, n at the 7 mm x-height, ascenders to the 12 mm line, descenders to the 6 mm line.

CHECK Ten consecutive letters sit on the baseline and touch the x-height line without drifting.

05 Set your stroke weights

Write "Come, palindove," at working size. Lay the printed stroke gauge beside the thinnest hairline: it must be at least 1 mm wide. Thick strokes should be 2-3 mm. If your hairlines are thinner, increase the minimum pressure or switch to the 1.0 mm marker for the master.

CHECK No stroke in the test line is narrower than the 1 mm gauge bar.

06 Practise the hard words

Use the hard-words sheet. Write each compound word as one word with no gap and no hyphen. Then practise the capitals that appear in the poem (C, D, E, F, I, L, O, S, T, shown darker on the alphabet sheet) with the flourish you intend to use for each, staying inside the dotted flourish lines.

CHECK Every flourish stays between the dotted ceiling and floor lines; nothing crosses into the next row.

07 Write the master rows in order

Use blank ruled sheets. One row per labelled row ID (R01 to R17, with a/b halves), in poem order, writing the row ID in pencil in the left label box. Keep one steady word gap of about 5-7 mm. Never split a word across rows. If a row runs long, stop at a word gap, write the rest on the next row, and label both with the same ID plus a and b.

CHECK Each row starts at the left rule and ends before the right margin; the text matches the exact-text sheet.

08 Let every sheet dry fully

Brush-pen pigment is touch dry in a minute; sumi needs an hour. Do not stack sheets or erase before then.

CHECK A clean tissue pressed on the darkest stroke picks up nothing.

09 Review with the checklist

Go row by row with the quick-reference checklist: wording, capitals, punctuation, x-height, slant, joins, stroke minimum, flourish zones, nothing touching the row above or below.

CHECK Every row has a pencil tick in its label box.

10 Rewrite, do not patch

If a row has a wrong letter, a blob, a stroke that touches another row, or a hairline under 1 mm, rewrite the whole row on a fresh ruled sheet with the same ID. Put a single pencil X through the rejected row. Do not use correction fluid on a master row; a clean rewrite scans cleaner than a repair.

CHECK Exactly one un-crossed row exists for each row ID.

11 Scan or photograph every sheet

Flatbed at 600 dpi, greyscale, no auto-contrast or sharpening, saved as TIFF or PNG named by sheet (for example jill-sheet-03.tif). Keep the whole sheet in the scan, including the row labels and the 100 mm bar, so scale and identity are recorded. If you must photograph: daylight, straight on, the whole sheet in frame, no shadows.

CHECK The scanned bar measures 100 mm when the file is opened at 600 dpi.

12 Approve the proof

Andrew traces the scans into outlines, joins the rows into the continuous ribbon, and sends back a proof at 1:1 and at 4:1 alongside a maquette-size print. Mark anything you want changed. Nothing is laser-marked before you approve the proof, and the first laser job is a small test coupon, not the full skin.

CHECK You have written “approved” and the date on the proof.

Keep the originals

Please keep the paper sheets flat and unsigned until the proof is approved. A signature or note can be added afterwards on a separate sheet if you would like it recorded with the archive.

Templates and reference render

Ruled sheets at the working size, ghosted reference rows rendered in the stand-in font, the alphabet, the hard words, and a true-size preview of the ribbon.

Every sheet is US Letter, landscape, with a 100 mm check bar in the header. Solid lines are baseline and x-height; long dashes are ascender and capital height; short dashes mark the descender line; dotted lines are the flourish ceiling and floor. Light diagonal lines show the 55° slant every 20 mm.

The 23 rows in poem order, three per sheet. Six long lines are written as two rows (a and b) and rejoined digitally.

SHEET 1

- R01** Come, palindove, let
edges twine;
- R02** thy side is mine, and mine
is thine.
- R03** Draw, O coward! cries the
knight.

SHEET 2

- R04** So inkhands draw him at
halfheight.
- R05** That halfknight draws a
quarterknight,
- R06** who draws a knightling,
out of sight.

SHEET 3

- R07** This swordfeud, quoth
one, is accursed:
- R08** each little inkprick's like
the first!
- R09a** Take sides! they roar.

SHEET 4

- R09b** The turncoat sighs:
- R10** I have but one. Do thou
likewise.
- R11a** Split it, then!

SHEET 5

- R11b** Slit it lengthwise through:
- R12** it comes round
doublelong, not two.
- R13a** Then who's our foe?

SHEET 6

- R13b** Thy mirrorknave.
- R14a** Look edgewise:
- R14b** that's thine own
swordwave.

SHEET 7

- R15a** Onesided! End it!
- R15b** The strip shrugs: Fine.
- R16a** Tail dovetails head;

SHEET 8

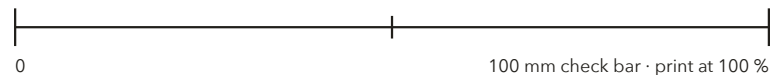
- R16b** mine's thine, thine's mine:
- R17** come, palindove, let
edges twine...

TOTAL

23 rows
on 8 sheets · writing width 235 mm

MUCH ADO ABOUT ONE SIDE • REFERENCE RENDER

Rows R01 to R03 in the stand-in font at the working size: aim for this size and weight in your own hand.



R01	<i>Come, palindove, let edges twine;</i>	flourish ceiling +22
		capitals +16
		ascender +12
		x-height +7
		baseline 0
		descender -6
		flourish floor -16
R02	<i>thy side is mine, and mine is thine.</i>	flourish ceiling +22
		capitals +16
		ascender +12
		x-height +7
		baseline 0
		descender -6
		flourish floor -16
R03	<i>Draw, O coward! cries the knight.</i>	flourish ceiling +22
		capitals +16
		ascender +12
		x-height +7
		baseline 0
		descender -6
		flourish floor -16

The same lettering at maquette size (2.5 mm em), actual size when printed at 100 %. This is why hairlines matter.

Come, palindove, let edges twine: thy side is mine, and mine is thine. Draw, O coward! cries the knight. So inbonds draw him at halfheight. That halffnighit draws a quaterknighit, who draws a knighitling, out of sight. This swordfeud, quoth one, is accursed:

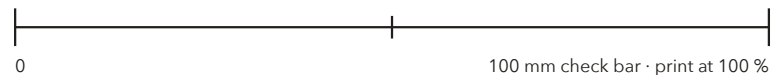
each little indprick's like the first! Take sides! they roar. The turncoat sighs: I have but one. Do thou likewise. Split it, then! Sht it longflurise through: it comes round doublelong, not two. Then who's our foe? Thy mirrorbrave.

Look edgewise: that's thine own swordweave. Oncsided! End it! The strip skruags: Fine. Tail dovetails head: mine's thine, thine's mine: come, palindove, let edges twine...

2.5 mm em · 10 mm ribbon band · horizontal scale 0.906 · actual size at 100 %

MUCH ADO ABOUT ONE SIDE • MASTER SHEET

Blank ruled sheet · x-height 7 mm · slant 55° · write one row ID per box



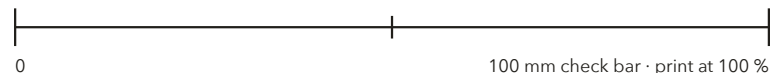
The image displays three identical sets of handwriting practice lines, each preceded by a small, empty square box. The lines are defined by the following vertical labels on the right side:

- flourish ceiling +22
- capitals +16
- ascender +12
- x-height +7
- baseline 0
- descender -6
- flourish floor -16
- flourish ceiling +22
- capitals +16
- ascender +12
- x-height +7
- baseline 0
- descender -6
- flourish floor -16
- flourish ceiling +22
- capitals +16
- ascender +12
- x-height +7
- baseline 0
- descender -6
- flourish floor -16

Solid: baseline and x-height. Long dash: ascender and capital. Short dash: descender. Dotted: flourish limits. Keep every stroke inside the dotted lines.

MUCH ADO ABOUT ONE SIDE • ALPHABET WARM-UP

Lowercase joined, then capitals. The poem's capitals (C D E F I L O S T) are shown darker.



flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6

flourish floor -16
flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6

flourish floor -16
flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6

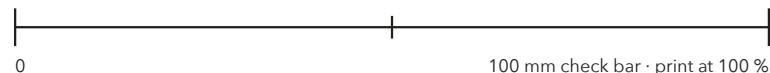
flourish floor -16
flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6

flourish floor -16

Write the lowercase alphabet twice and each capital once, joined as in running text. Check ten letters in a row sit on the baseline and touch the x-height line.

MUCH ADO ABOUT ONE SIDE • HARD WORDS

Invented and compound words, each written as one word with no gap or hyphen.



flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6

palindove inkhands halfheight halfknight

flourish floor -16
flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6

quarterknight knightling swordfeud

flourish floor -16
flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6

inkprick's doublelong swordwave

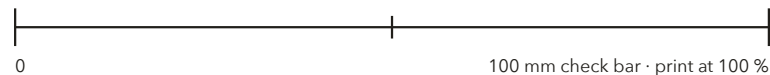
flourish floor -16
flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6

mirrorknave Onesided

flourish floor -16

MUCH ADO ABOUT ONE SIDE • REFERENCE SHEET 1 OF 8

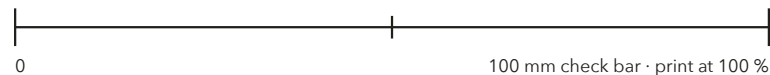
Rows R01 to R03 · stand-in font at working size · size and spacing reference, not letterforms



R01	<i>Come, palindove, let edges twine;</i>	flourish ceiling +22
		capitals +16
		ascender +12
		x-height +7
		baseline 0
R02	<i>thy side is mine, and mine is thine.</i>	descender -6
		flourish floor -16
		flourish ceiling +22
		capitals +16
		ascender +12
R03	<i>Draw, O coward! cries the knight.</i>	x-height +7
		baseline 0
		descender -6
		flourish floor -16
		flourish ceiling +22

MUCH ADO ABOUT ONE SIDE • REFERENCE SHEET 2 OF 8

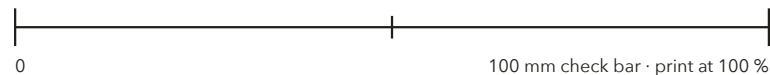
Rows R04 to R06 · stand-in font at working size · size and spacing reference, not letterforms



R04	<i>So inkhands draw him at halfheight.</i>	flourish ceiling +22
		capitals +16
		ascender +12
		x-height +7
		baseline 0
		descender -6
		flourish floor -16
R05	<i>That halfknight draws a quarterknight,</i>	flourish ceiling +22
		capitals +16
		ascender +12
		x-height +7
		baseline 0
		descender -6
		flourish floor -16
R06	<i>who draws a knightling, out of sight.</i>	flourish ceiling +22
		capitals +16
		ascender +12
		x-height +7
		baseline 0
		descender -6
		flourish floor -16

MUCH ADO ABOUT ONE SIDE • REFERENCE SHEET 3 OF 8

Rows R07 to R09a • stand-in font at working size • size and spacing reference, not letterforms



R07 *This swordfeud, quoth one, is accursed:*

flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6
flourish floor -16

R08 *each little inkprick's like the first!*

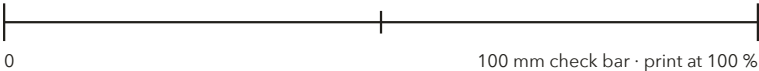
flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6
flourish floor -16

R09a *Take sides! they roar.*

flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6
flourish floor -16

MUCH ADO ABOUT ONE SIDE • REFERENCE SHEET 4 OF 8

Rows R09b to R11a · stand-in font at working size · size and spacing reference, not letterforms



R09b

The turncoat sighs:

flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6
flourish floor -16

R10

I have but one. Do thou likewise.

flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6
flourish floor -16

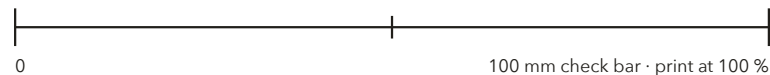
R11a

Split it, then!

flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6
flourish floor -16

MUCH ADO ABOUT ONE SIDE • REFERENCE SHEET 5 OF 8

Rows R11b to R13a · stand-in font at working size · size and spacing reference, not letterforms



R11b

Slit it lengthwise through:

flourish ceiling +22

capitals +16

ascender +12

x-height +7

baseline 0

descender -6

flourish floor -16

R12

it comes round doublelong, not two.

flourish ceiling +22

capitals +16

ascender +12

x-height +7

baseline 0

descender -6

flourish floor -16

R13a

Then who's our foe?

flourish ceiling +22

capitals +16

ascender +12

x-height +7

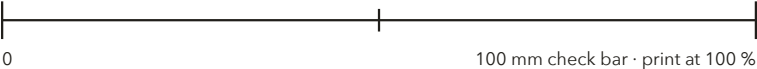
baseline 0

descender -6

flourish floor -16

MUCH ADO ABOUT ONE SIDE • REFERENCE SHEET 6 OF 8

Rows R13b to R14b • stand-in font at working size • size and spacing reference, not letterforms



R13b

Thy mirrorknave.

flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6
flourish floor -16

R14a

Look edgewise:

flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6
flourish floor -16

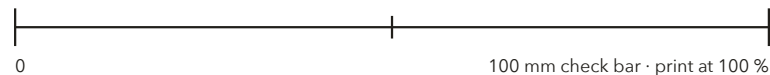
R14b

that's thine own swordwave.

flourish ceiling +22
capitals +16
ascender +12
x-height +7
baseline 0
descender -6
flourish floor -16

MUCH ADO ABOUT ONE SIDE • REFERENCE SHEET 7 OF 8

Rows R15a to R16a • stand-in font at working size • size and spacing reference, not letterforms



R15a

Onesided! End it!

flourish ceiling +22

capitals +16

ascender +12

x-height +7

baseline 0

descender -6

flourish floor -16

R15b

The strip shrugs: Fine.

flourish ceiling +22

capitals +16

ascender +12

x-height +7

baseline 0

descender -6

flourish floor -16

R16a

Tail dovetails head;

flourish ceiling +22

capitals +16

ascender +12

x-height +7

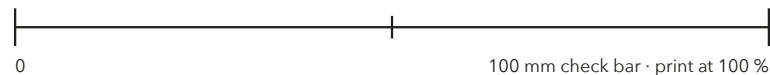
baseline 0

descender -6

flourish floor -16

MUCH ADO ABOUT ONE SIDE • REFERENCE SHEET 8 OF 8

Rows R16b to R17 • stand-in font at working size • size and spacing reference, not letterforms



R16b

mine's thine, thine's mine:

flourish ceiling +22

capitals +16

ascender +12

x-height +7

baseline 0

descender -6

flourish floor -16

R17

come, palindove, let edges twine...

flourish ceiling +22

capitals +16

ascender +12

x-height +7

baseline 0

descender -6

flourish floor -16

flourish ceiling +22

capitals +16

ascender +12

x-height +7

baseline 0

descender -6

flourish floor -16

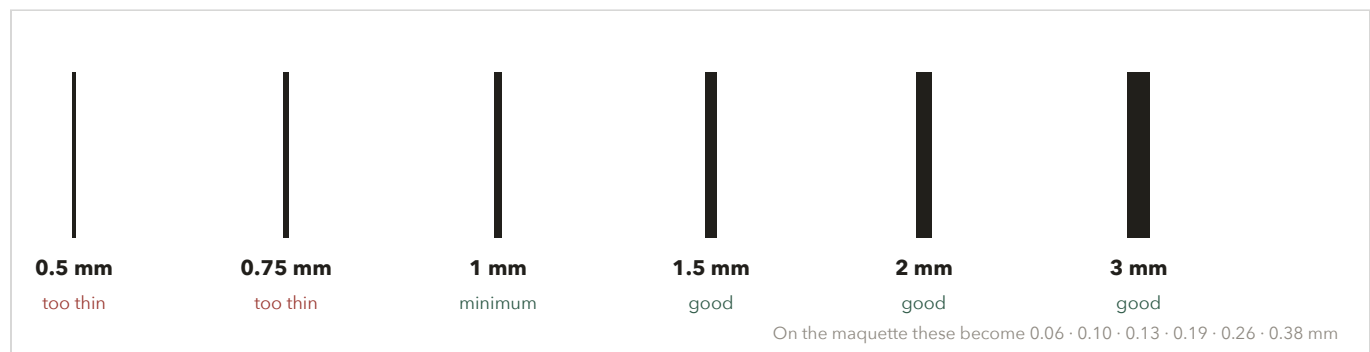
The easy parts

If you only read one page, read this one. Sizes, the stroke rule, the checklist, and what to send back.

SIZES AT THE WORKING SCALE

ELEMENT	ON YOUR PAPER	ON THE 180 MM MAQUETTE	IN EMS
x-height (body of a, o, n)	7 mm	0.89 mm	0.36
Em (row size)	19.6 mm	2.5 mm	1.00
Ascender (h, l, d, k)	12 mm	1.53 mm	0.62
Capital height	16 mm	2.04 mm	0.81
Descender (p, g, y, f)	6 mm	0.77 mm	0.32
Flourish ceiling (above baseline)	22 mm	2.81 mm	1.12
Flourish floor (below baseline)	16 mm	2.04 mm	0.80
Row pitch	48 mm	6.12 mm	2.4
Thinnest stroke, minimum	1 mm	0.13 mm	0.05
Thick strokes	2-3 mm	0.26 mm-0.38 mm	0.10-0.15
Gap between words	5-7 mm	0.64 mm-0.89 mm	0.25-0.35
Slant from the baseline	55°	55°	—

Stroke gauge. Hold a written hairline against the bars: it must be at least as wide as the 1 mm bar.



Do

- Write in your own slanted, joined script with flourished capitals.
- Keep the x-height on the 7 mm line and the slant near 55°.

- Keep the thinnest stroke at 1 mm or wider.
- Keep flourishes inside the dotted lines.
- Write compound words as single words: palindove, inkhands, halfknight.
- Copy punctuation exactly, including the final ellipsis.
- Label every row; rewrite rather than patch.

Don't

- Don't add words, hyphens, or extra capitals.
- Don't split a word across two rows.
- Don't let any stroke touch the row above or below.
- Don't use coloured, glossy, or metallic ink.
- Don't crop the row labels or the 100 mm bar out of the scans.

Row checklist

- | | |
|--|--|
| ☐ Wording matches the exact-text sheet, letter for letter | ☐ Slant consistent along the row |
| ☐ Capitals only where the poem has them | ☐ Thinnest stroke at least 1 mm |
| ☐ Punctuation present and correct | ☐ Flourishes stay inside the dotted lines |
| ☐ Bodies of letters sit on the baseline and reach the x-height line | ☐ Row ID written in the label box |

What to send

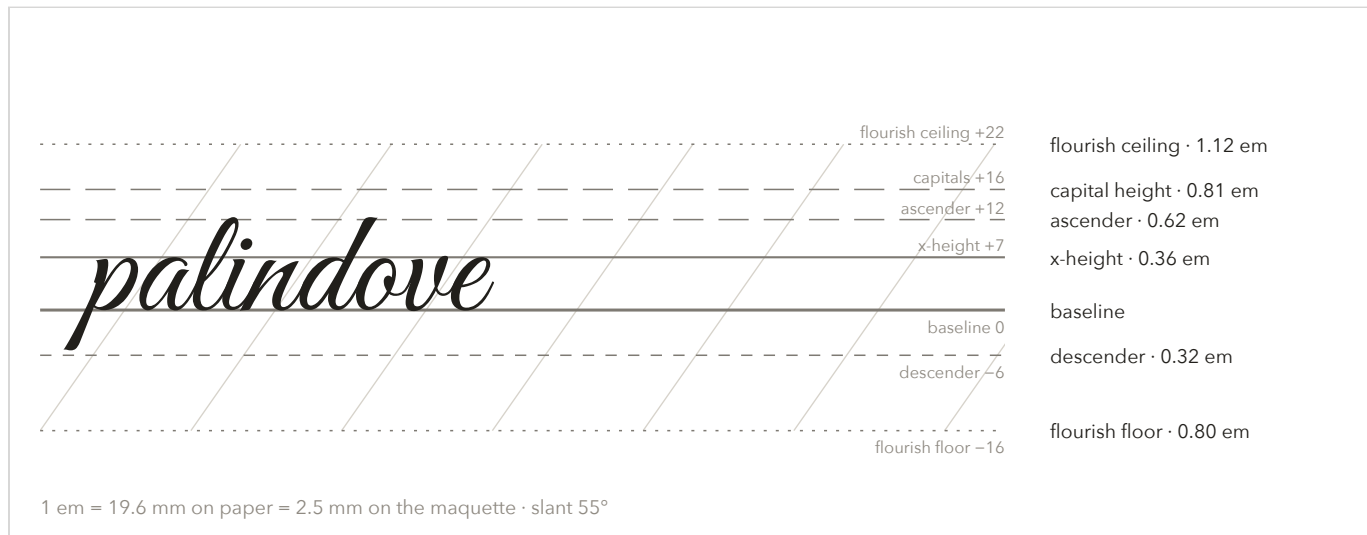
1. 23 approved rows across the sheets, scanned at 600 dpi as TIFF or PNG, one file per sheet, with the full sheet in frame.
2. One photo of the pen you used and the paper's packaging, so the archive records the materials.
3. Any notes on rows you were unsure about.

In depth

Proportions, spacing and joins, flourish zones, what the laser can and cannot mark, materials in detail, how the scan becomes foil, and what to do when things go wrong.

Proportions

The guideline set on every sheet, with the stand-in font's "palindove" placed on it for scale.



The layout is built around the em, the nominal row size. The stand-in font has an x-height of 0.36 em, ascenders at 0.62 em, capitals at 0.81 em, and descenders to 0.32 em. At the working size that gives a 7 mm x-height, 12 mm ascenders, 16 mm capitals and 6 mm descenders. Your natural proportions may differ a little; what matters is that they stay the same on every row, because rows are scaled to a common x-height when they are joined.

The working sheets and the jaw master space rows 2.4 ems apart, which is 48 mm at the working size. The dotted flourish lines mark the space a letter may use without touching a neighbouring row: up to 22 mm above the baseline and 16 mm below. The maquette's ribbon band is 10 mm tall for a 2.5 mm em, so the same limits apply there. The large body master currently stacks 4 rows only 1.4 ems apart; before that run Andrew re-spaces the body rows to the real extent of your capitals and descenders, so do not tighten your hand for it.

Spacing, joins, and slant

- Letters within a word join in the usual running-script way; keep the spacing between joined letters even and fairly tight, roughly the width of a lowercase n between stems.
- Words are separated by one steady gap of 5–7 mm, about one x-height. The gap between poem lines on the ribbon is doubled (10–14 mm); Andrew adds that when the rows are joined, so end each row after the final punctuation with no special gap.
- Slant is 55° from the baseline, the classic copperplate angle, drawn faintly on every sheet. If your natural slant is a few degrees different, use yours and keep it constant.
- The ribbon is one continuous line, so the tail of each row must be able to meet the head of the next at a word gap. Exit strokes may trail a little; an entry flourish on a capital may lead in a little; neither should be longer than one word gap.
- The final ellipsis runs into the first word. Give the three dots a normal ending and let the first “C” carry whatever lead-in flourish you like; the join will read as a loop either way.

Capitals and flourishes

The poem uses these capitals: C, D, E, F, I, L, O, S, T. Each begins a line or a sentence except the lowercase “come” that starts the last line, which is deliberate: the poem loops, so its last line is a continuation. Flourished capitals are welcome. Keep flourishes clear of the words before and after them, and never above the ceiling or below the floor line. A flourish that crosses another letter becomes an unreadable blob at 2.5 mm.

Why the 1 mm stroke rule

The foil is 0.127 mm AlumaMark aluminum. A CO₂ laser darkens its coating rather than cutting it, and the marked line cannot be narrower than the beam's focused spot, roughly 0.1–0.2 mm. Your master is scaled by 0.128 to reach the maquette's 2.5 mm em, so a 1 mm stroke on paper becomes 0.13 mm on the foil, right at the limit of what survives. Anything thinner drops out or fills in. Thick strokes of 2–3 mm keep the script's contrast without closing up the counters of e, a, and o.

If you prefer a pointed dip pen

Copperplate hairlines from a Nikko G or Zebra G nib are 0.1–0.2 mm on paper and would vanish. If that is the hand you want, write it anyway and say so: Andrew will thicken every outline uniformly by about 0.4 mm on the paper scale before scaling. The hairlines then survive, but the contrast softens and the letters gain a little weight. A brush pen avoids that step entirely.

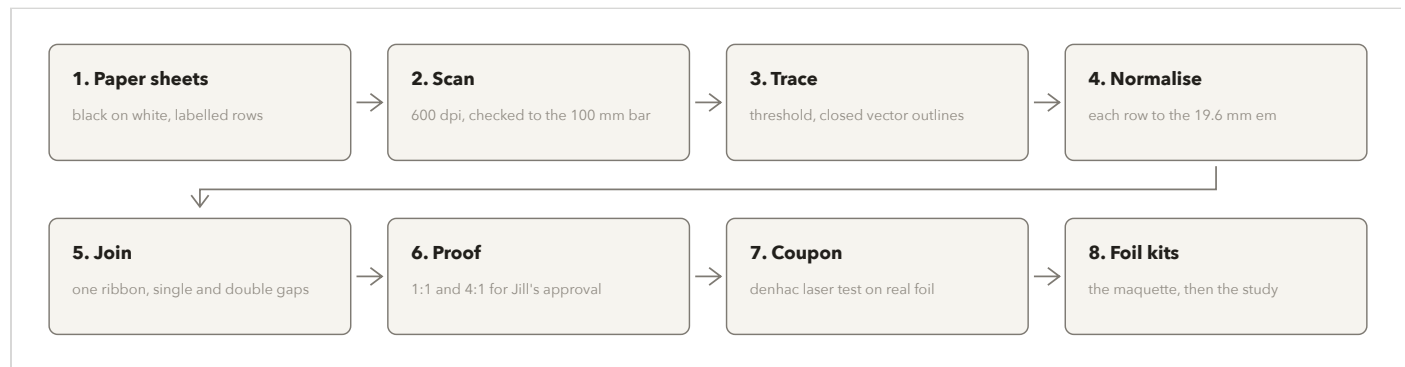
Materials in detail

ITEM	USE THIS	ALSO FINE	AVOID
Pen	Pointed brush pen with a black pigment cartridge: Pentel Pocket Brush (GFKP3) or Kuretake No. 13. Keep the lightest strokes at 1 mm or more.	A 1.0 mm round-tip pigment marker (Sakura Pigma Graphic 1, Staedtler pigment liner 1.0) for a monoline master.	Pointed dip nibs and broad-edge pens on their own: their hairlines are 0.1–0.4 mm and vanish at maquette size unless we thicken them digitally (see In depth).
Ink	The pen's own black pigment cartridge.	For dip pens: Moon Palace or Kuretake sumi, Higgins Black Magic, Dr. Ph. Martin's Black Star matte.	Blue-black, grey, walnut, iron-gall, gel pens with sheen, metallic or glossy inks, ballpoint, pencil.
Paper	Smooth, bright white, 120–270 g/m ² : HP Premium Choice 32 lb, Clairefontaine 120 g, or Strathmore 400 smooth Bristol.	Rhodia or Clairefontaine 90 g pads for brush pens; marker layout paper for practice.	Textured or cotton watercolour paper, cream or toned stock, lined notebook paper, thin copier paper.
Guidelines	Print the template sheets at 100 % straight onto the writing paper.	Print on plain paper and place it under Bristol on a light pad or window, taped with low-tack tape.	Freehand guessing at x-height; guidelines drawn in ink.
Tools	Steel ruler with millimetres, 2H pencil, plastic eraser, low-tack (washi) tape, scrap of the same paper for pen tests, tissue.	Light pad, T-square, a 55° slant card cut from the template.	Correction fluid or white gouache patches on the master rows.
Capture	Flatbed scan at 600 dpi, greyscale, no auto-enhance, saved as TIFF or PNG.	A phone photo in daylight, straight on, whole sheet in frame including the 100 mm bar, if a scanner is impossible.	JPEG at low quality, cropped scans that lose the row labels or scale bar, skewed or shadowed photos.

Matte black on smooth bright white scans as a clean two-tone image, which is what vector tracing needs. Glossy inks reflect the scanner lamp and produce grey speckles inside strokes; textured paper adds noise along every edge; cream paper lowers contrast. Bristol or a heavy laser paper also stays flat on the scanner glass.

From scan to foil

What happens to the sheets after they are scanned.



1. Each scan is checked against its 100 mm bar and straightened using the printed baselines.
2. The image is thresholded to pure black and white and traced into closed vector outlines, the same kind of outlines the current stand-in font produces.
3. Each row is measured from its baseline and x-height lines and normalised to the 19.6 mm em, so rows written on different days still match.
4. Rows are joined in poem order with one word gap inside a line and a double gap between lines, producing one continuous ribbon.
5. The ribbon's length is compared with the stand-in layout (about 194 ems for one poem). Up to about 15 % difference is absorbed by adjusting word gaps and a small uniform scale; outside that we talk before changing anything.
6. The ribbon replaces the font outlines in the generators for both editions: two repetitions around the maquette's 1028 mm body circuit, and one poem per row on the large study.
7. You receive a proof. After approval, denhac staff mark a small test coupon with your lettering at 2.5, 4, 6, 8, 10 and 14 mm em sizes on the actual foil, and we look at it together before any full sheet is marked.

Troubleshooting

PROBLEM	LIKELY CAUSE	FIX
Edges feather or bleed	Absorbent or textured paper, or a wet dip-pen line	Switch to smooth Bristol or 120 g laser paper; let the pen's own cartridge ink do the work.
Hairlines under 1 mm	Too little pressure on the up-strokes	Slow down on up-strokes and keep the brush tip slightly loaded; or use the 1.0 mm marker for the master.
Counters of e, a, o fill in	Thick strokes over 3 mm at this size	Ease the pressure on down-strokes; the gauge's 3 mm bar is the ceiling.
x-height drifts along the row	Writing without looking at the line, or fatigue	Write in shorter bursts; check the height every three words; rewrite the row if it drifts more than 1 mm.
Slant wanders	Paper rotated, or guidelines too faint	Tape the sheet square to the table; darken the slant lines with pencil if needed.
Row runs out of room	Wide hand or generous word gaps	Stop at a word gap and continue on the next row as part b; never squeeze the last word.
Ink pooling at joins	Pausing with the tip on the paper	Lift between words; keep the pen moving through joins.
Scan looks grey or speckled	Auto-enhance, glossy ink, or low resolution	Rescan at 600 dpi greyscale with all automatic corrections off.

Punctuation inventory

MARK	NAME	COUNT
,	comma	13
;	semicolon	2
:	colon	6
!	exclamation mark	6
?	question mark	1
.	full stop	11
'	apostrophe	5
...	ellipsis (one character, three dots)	1

The apostrophes belong to inkprick's, who's, that's, mine's and thine's. The only ellipsis is the last character of the poem. There are no quotation marks, dashes, or parentheses anywhere.